BATON Research Continuity Platform (Round 2 Submission)

Team Kernel Connect -> Reconstruct -> Verify -> Review -> Handover Verified: 2026-11-11

1. What Changed Since Round 1 (Proposal -> Working Full-Stack Product)

Round 1 Baseline: Conceptual design proposal only (Stage 00 audit confirmed zero application code in workspace).
Round 2 Shipped Implementation: Full-stack Node.js v22 + relational SQLite application (7 SQL migrations, 19 tables) and a 7-screen Vanilla HTML/CSS/JS workspace verified by 64/64 passing automated integration & E2E subtests.
Stage 1 (Connect): Multi-format upload (PDF, Markdown/Text, CSV, JSON), exact PDF_PAGE / NOTE_SECTION / CSV_ROW / JSON_RECORD span indexing with character offsets, SHA-256 duplicate rejection (409), scanned PDF detection (422 SCANNED_PDF_NEEDS_OCR), and untrusted prompt-injection pattern flagging & sandboxing.
Stage 2 (Reconstruct): Deterministic SQL pre-filtering of RESTRICTED_PI_ONLY sources before context construction, extraction of 11 continuity record types separating Direct Source Statement from AI Summary, null/UNKNOWN preservation for unstated owners/dates/rationale, and Continuity Map edges (CONFLICTS_WITH, EVIDENCE_CORROBORATES, ADDRESSES_GAP).
Stage 3 (Verify): 6 canonical evidence states (SUPPORTED, CONFLICTING, UNSUPPORTED, STALE, NEEDS REVIEW, NOT ESTABLISHED), zero fabricated confidence percentages, side-by-side conflicting passages, and citation flagging.
Stage 4 & 5 (Review & Handover): Outgoing researcher edits with field diffs, approval bound to record version, post-approval edit invalidation, 6-section Continuation Plan, and versioned recipient receipt acknowledgement.

2. Verified Deployment & Repository URLs (Honest Status Disclosure)

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Local Production Server URL : http://127.0.0.1:3000  (Verified live: GET /api/health -> 200 OK, 7 migrations)
External Public Cloud URL   : NOT DEPLOYED (Built & verified inside offline local sandbox environment;
                              turnkey Dockerfile and render.yaml with /var/data persistent disk included).
GitHub Repository URL       : https://github.com/pournima-d/baton-research-continuity
Safe Demo Access Personas   : PI_LEAD (tok_pi_lead_01), RESEARCHER (tok_researcher_01), REVIEWER (tok_reviewer_01),
                              INCOMING_MEMBER (tok_incoming_01), AUDITOR (tok_auditor_01)   zero privileged secrets.
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3. System Architecture & Pre-AI Restricted Data Flow

```
[1. Connect: PDF / Note / CSV / JSON] ---> Private Storage (/storage/private_uploads) + SQLite (sources, source_spans)
       |
       +---> Deterministic SQL Pre-Filter: WHERE visibility_level = "PROJECT_SHARED" AND exclude_from_ai = 0
             (RESTRICTED_PI_ONLY sources are excluded BEFORE extraction, summarisation, or handover assembly)
       |
[2. Reconstruct: aiReconstructor.js] ----> Candidate Records (DRAFT) + Evidence Links + Continuity Map Edges
       |
[3. Verify: evidenceInspector.js] -------> SUPPORTED | CONFLICTING | UNSUPPORTED | STALE | NEEDS REVIEW | NOT ESTABLISHED
       |
[4. Review: reviews & record_versions] --> APPROVE (bound to version) | REQUEST_CHANGES | FLAG_GAP | REJECT
       |
[5. Handover: handoverService.js] -------> Published by PI_LEAD -> Role-Filtered View -> Receipt Acknowledged (v1)
```

BATON — Actual Product Workspace Captures & Shipped Workflow

Captured directly from running application state (docs/screenshots/01..03.svg)

Workspace Capture 1: Stage 00 Overview & Stage 01 Connect (docs/screenshots/01_overview_and_connect.svg)

Project: Solid-State Li-Garnet (LLZO) Electrolyte Interface Continuity (prj_solidstate_01) | Role: PI_LEAD
Metrics: 4 Ingested Sources (3 PROJECT_SHARED, 1 RESTRICTED_PI_ONLY) | 10 Provenance Spans | 3 Seeded Records

SOURCE ID	TYPE	VISIBILITY	STATUS	INDEXED PROVENANCE LOCATORS
src_pdf_01	PDF_DOCUMENT	PROJECT_SHARED	READY	4 spans (PDF_PAGE: Page 1..2, char offsets 0..762)
src_note_01	LAB_NOTE	PROJECT_SHARED	READY	2 spans (NOTE_SECTION: Argon Purge & Drift Log)
src_csv_01	CSV_DATASET	PROJECT_SHARED	READY	4 spans (CSV_ROW: Samples SSE-01 to SSE-04)
src_restricted	LAB_NOTE	RESTRICTED_PI_ONLY	READY	1 span (Excluded from AI & non-PI roles)

Sample Span [spn_pdf01_01] (Page 1, Paragraph 1): "Objective: Achieve >93% relative density and >0.60 mS/cm room-temperature ionic conductivity in Ta-doped Li7La3Zr2O12 (LLZO) solid electrolyte pellets..."

Workspace Capture 2: Stage 02 Continuity Map & Stage 03 Evidence Inspector (02_continuity_map_and_verify.svg)

Node [DECISION rec_seed_01]: Switch from Alumina to MgO Crucibles | Evidence: SUPPORTED | Approval: APPROVED (v1)

- Direct Source Statement : "Switched crucible setup from bare alumina to MgO crucibles with mother-powder..."
- AI Summary / Inference : Standardized MgO crucible protocol raised pellet density to 94.2% (0.68 mS/cm).
- Model Disclosure : provider=baton-local-deterministic | model=heuristic-span-reconstructor-v2 (v2.5.0)

Conflicting Pair Detected : Sample SSE-09 conductivity (2026-09-10: 0.72 mS/cm) vs (2026-09-18: 0.31 mS/cm)

- Edge: CONFLICTS_WITH | Both passages displayed side-by-side without choosing a winner automatically.

Stale Calibration Flagged : EQ-PRESSURE-01 (2026-07-01) marked STALE relative to 2026-10-15 recalibration log.

Missing Owner Preserved : Task TSK-102 owner=null, due_date=null -> flagged is_incomplete=1 (never fabricated).

Epistemic Citation Notice : "A citation shows where a claim came from; it does not prove the claim is true."

Accessible Map Toggle : #continuity-map-table-card provides a full semantic HTML table alternative.

Workspace Capture 3: Stage 04 Review, Stage 05 Handover & Stage 08 Audit Trail (03_review_handover_and_audit.svg)

Governance Invariant : Editing an APPROVED record creates v(N+1), resets to PENDING_REVIEW, invalidates approval.

Continuation Plan (v1) : Generated from APPROVED records only (restricted_sources_excluded_count: 1):

1. Current Status : 3 Approved Records | 0 Rejected/Draft | 1 Restricted PI-Only source excluded server-side
2. Approved Decisions : Switch to MgO Crucibles with Mother-Powder Bed (Approved v1 by Dr. Priya Nair)
3. Open Tasks & Next Steps: TSK-102 Owner: UNKNOWN (Not stated in source) | Deadline: UNKNOWN (No deadline stated)
4. Blockers, Risks & Gaps : High-current (>0.8 mA/cm2) interfacial voiding & H2O analyzer drift (+1.8 ppm)

Recipient Acknowledgement : Aria Patel (INCOMING_MEMBER) checked all items -> status: ACKNOWLEDGED (v1)

- Disclaimer: "Acknowledgement means receipt, not agreement or proof of understanding."

Append-Only Audit Events : PROJECT_CREATED, SOURCE_UPLOADED, UNTRUSTED_PROMPT_PATTERN_SANDBOXED, AI_CONTEXT_FILTERED, RECONSTRUCTION_COMPLETED, RECORD_VERIFIED, RECORD_EDITED, REVIEW_SUBMITTED, HANDOVER_CREATED, HANDOVER_PUBLISHED, HANDOVER_ACKNOWLEDGED, ACCESS_DENIED (Immutable via HTTP 405 & SQLite triggers).

4. Technical & AI Disclosures (DISCLOSURES.md)

Runtime & Database : Node.js v22.14.0 built-in modules (node:http, node:sqlite, node:crypto, node:test, node:vm). Zero third-party npm packages (0 external node_modules). SQLite 3.x with WAL & FK enforcement.

Extraction Engine : First-party deterministic structured span reconstructor (baton-local-deterministic / heuristic-span-reconstructor-v2 v2.5.0); optional GEMINI_API_KEY (gemini-2.5-flash) config hook.

UI & Templates : 100% custom-authored HTML5, CSS3, and Vanilla ES6 JS (public/). No third-party UI templates.

AI Coding Assistant: Google DeepMind Antigravity used for pair-programming implementation & test execution.

5. Automated Test Suites (64/64 PASS) & Stage 08 Security Test Matrix (TEST_REPORT.md)

SUITE / SECURITY MATRIX ITEM	OBSERVED RESULT & EVIDENCE	
1. Unauthorised project access	PASS	401 UNAUTHENTICATED & 403 FORBIDDEN_ROLE (tok_outsider_01)
2. Collaborator attempts to approve	PASS	RESEARCHER blocked (403 FORBIDDEN_ROLE / APPROVAL_BYPASS)
3. Recipient requests restricted source	PASS	403 RESTRICTED_RESOURCE on src_restricted_01 (/spans, /file)
4. Restricted text absent from AI input	PASS	SQL pre-filter excludes restricted sources; 0 canary leakage
5. Guessed source/file ID & direct /storage/*	PASS	404 RESOURCE_NOT_FOUND & 403 FORBIDDEN_STORAGE_PATH
6. Cross-project evidence ID	PASS	422 CROSS_PROJECT_EVIDENCE_VIOLATION across all endpoints
7. Malformed, empty, oversized & scanned uploads	PASS	400/413/422; scanned PDF marked NEEDS_OCR honestly
8. Prompt-injection text inside source	PASS	Flagged (has_prompt_injection_pattern=1), stripped, DRAFT only
9. Audit event generated & append-only immutability	PASS	All denials logged; HTTP 405 & SQLite triggers block edits
10. Sessions, scrypt, CSRF, CORS, Rate Limits	PASS	HttpOnly/SameSite=Strict/Secure, 403 CSRF/CORS, 429 rate limit
11. DB Restart Persistence & Model Timeout (504)	PASS	State persists across DB close/reopen; 0 partial records on 504

6. Honest Stage 09 Camera Step-Up Status

Optional & Risk-Triggered: Implemented for sensitive actions (PUBLISH_HANDOVER_PACKAGE, ACCESS_RESTRICTED_SOURCE) with a first-class, equivalent Non-Camera Route (NON_CAMERA_MFA_PASSKEY) as the recommended default.

Honest Capability Disclaimer: Browser camera permission only means a local stream was made available. It does NOT perform facial recognition, verify identity, prove who is present, or prove liveness. The authenticated session and MFA/passkey verification remain the actual security control (camera stream without valid MFA is rejected with 401).

Zero Media / Biometric Storage: Enforced by SQLite CHECK (media_stored = 0) and 400 MEDIA_UPLOAD_PROHIBITED.

All Error States Tested: PERMISSION_DENIED, NO_DEVICE, UNSUPPORTED_BROWSER, CANCELLED, and DEVICE_ERROR tested in tests/camera_step_up_stage09.test.js (6/6 PASS). Physical webcam capture in a GUI browser is marked NOT VERIFIED.

7. Documented Security & Product Limitations (SECURITY_NOTES.md)

Local Disk Storage: SQLite (data/baton.db) and private blobs are unencrypted at rest on the host OS filesystem.

In-Memory Rate Limiter: Rate limit counters reside in process memory; multi-node scaling requires Redis.

Heuristic Prompt-Injection Stripping: Strips override/tool patterns, but human verification & review remain mandatory.

Scanned PDF OCR: Scanned PDFs without text streams are flagged NEEDS_OCR; external OCR is not bundled in Round 2.

8. Three Round 3 Commitments (Not Claimed as Shipped Round 2 Features)

A. Knowledge-Debt Radar:
Identify critical project knowledge concentrated in a single person (stated_owner concentration) or weakly documented (UNSUPPORTED, NEEDS REVIEW, STALE, or missing rationale/owner), producing prioritised gap alerts.

B. Cold-Start Handover Test:
Practical newcomer questions test whether the approved handover record can answer essential operational and experimental questions reporting record and documentation gaps in the package, never a person's competence.

C. Sensitive-Content Detection and Redaction:
Detect likely email addresses, keys/tokens, and personal data in uploaded sources and draft records, suggesting redactions for explicit human confirmation before sharing or bundling into a handover package.

9. Actual Team Contributions (CONTRIBUTIONS.md)

Team Kernel (Pournima Project Lead & Architect):
Formulated the Round 1 research continuity problem and 5-stage governance workflow; directed the 12-stage Round 2 build, security architecture, epistemic evidence rules, and stage-by-stage verification acceptance checks.
AI Pair-Programming Assistant (Google DeepMind Antigravity):
Paired with the user to implement the Node.js/SQLite backend, 7 migrations, 7-screen frontend workspace, 9 automated test suites (64/64 passing), security controls, and Round 2 documentation.

10. Judge-Facing Three-Minute Demo Script (Verified Working Features Only)

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0:00 - 0:45 | Stage 1 (Connect) & Role-Based Source Visibility:
1. Open http://127.0.0.1:3000 as RESEARCHER (Marcus Chen). Show live SQLite metrics on Overview.
2. Open "01 Sources": inspect 3 PROJECT_SHARED sources (PDF, Markdown Note, CSV). Switch role to PI_LEAD
   (Dr. Elena Vance) to reveal "src_restricted_01" (RESTRICTED_PI_ONLY).
3. Click "Inspect Spans" on llzo_sintering_report.pdf to show exact PDF_PAGE locators and character offsets.

0:45 - 1:45 | Stage 2 (Reconstruct) & Stage 3 (Verify Evidence & Gaps):
4. Open "02 Continuity Map" as RESEARCHER and run Pre-Filtered Reconstruction. Point out the filter summary:
   "Restricted/AI-Excluded Sources Filtered Out: 1" (restricted text never entered extraction context).
5. Toggle "Table View (Accessible)" to show the keyboard/screen-reader table alternative and Continuity Edges.
6. Open "03 Records & Evidence": inspect separated Direct Source Statement vs AI Summary, model disclosure,
   and the 6 canonical evidence states. Click "Flag Incorrect Citation" to see automatic downgrade from SUPPORTED.

1:45 - 3:00 | Stage 4 (Review), Stage 5 (Handover) & Append-Only Audit Trail:
7. Switch to REVIEWER (Dr. Priya Nair) in "04 Review Queue" and APPROVE a record (bound to version, preserving
   evidence_status). Show that editing an approved record resets it to PENDING_REVIEW.
8. In "05 Handover", publish as PI_LEAD (showing optional Non-Camera MFA Step-Up), then switch to INCOMING_MEMBER
   (Aria Patel), verify restricted content is absent, check all items, and click "Acknowledge Receipt".
9. Switch to AUDITOR (Jordan Lee) in "06 Activity / Audit" to inspect the immutable audit_events trail.
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